

Task 12

Step 1: Create EC2 with Auto Scaling Group

1. Create a **Launch Template**:
 - a. AMI: Amazon Linux 2
 - b. Instance type: t2.micro (or equivalent)
 - c. Attach an **IAM Role** with:
 - i. CloudWatch access
 - ii. EC2 access (read/write)
 - iii. SNS publish permission
 - d. Install a simple web server (Apache/Nginx)
2. Create an **Auto Scaling Group (ASG)**:

This ensures EC2 can be replaced automatically.

Step 2: Configure CloudWatch Monitoring

1. Go to **CloudWatch → Alarms**
2. Create an alarm based on **EC2 metric**:

This alarm represents “failure detection”.

Step 3: Create SNS Topic for Notifications

1. Go to **SNS → Topics**
2. Create a standard topic (example: ec2-self-healing-alerts)
3. Add email subscription(s)
4. Confirm the subscription from your email inbox

SNS will notify when failure and recovery occur.

Step 4: Create Lambda Function (Self-Healing Logic)

Lambda is the **automated fixer**.

Step 5: Connect CloudWatch Alarm to Lambda

1. Edit the **CloudWatch Alarm**
2. Set alarm action:

- a. Trigger the **Lambda function**
3. Save the alarm

This connects detection → action.

Step 6: Intentionally Fail the EC2 Instance

Use **any ONE** of the following:

Option 1 (Recommended):

- Stop the EC2 instance from AWS Console
Do NOT manually recover the instance.

Step 7: Observe Self-Healing Behavior

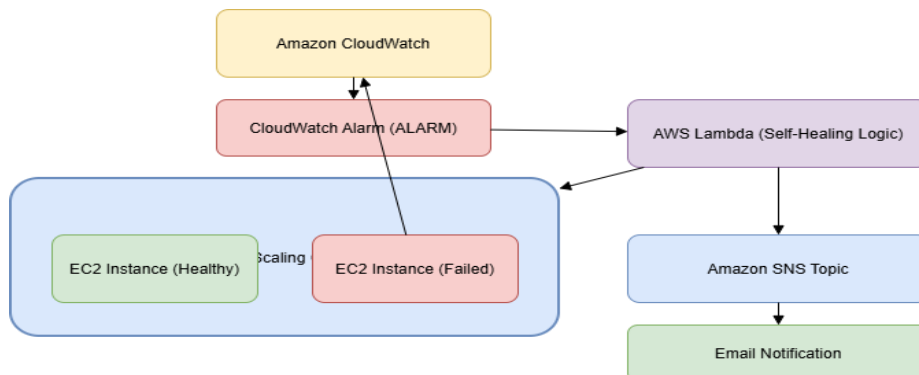
Verify the following:

- CloudWatch alarm changes to **ALARM**
- Lambda function is invoked
- Auto Scaling replaces or recovers EC2
- SNS notification email is received
- New EC2 instance becomes healthy

Output:

EC2 failure must be detected, fixed, and notified **automatically**, without any manual intervention.

Workflow:



Solution:

1. Login to your AWS account with credentials and required verification.

(<https://signin.aws.amazon.com/>)

My account – AWS Management Console- Sign in

aws

Sign In

Access your AWS account by user type.

User type [\(not sure?\)](#)

☒ Root user
Account owner that performs tasks requiring unrestricted access.

☐ IAM user
User within an account that performs daily tasks.

Email address

Build your own frontier models
Nova Forge offers the easiest and most cost-effective way to build your own frontier models.
[Try Nova Forge](#)

Create Auto Scaling Group

2. For Auto Scaling we need to create an IAM Role

- Go to IAM - Roles - Create role
- Trusted entity:
 - Select AWS service
 - Choose EC2
- Attach policies:
 - *CloudWatchAgentServerPolicy*
 - *AmazonEC2FullAccess*
 - *AmazonSNSFullAccess*
- name: *Ec2-SelfHealing*
- Create role.

Step 1: Select trusted entity

Step 2: Add permissions

Step 3: Name, review, and create

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

EC2

Choose a use case for the specified service.

Use case

- ☒ **EC2**
Allows EC2 instances to call AWS services on your behalf.

Step 1: Select trusted entity

Step 2: Add permissions

Step 3: Add tags

Permissions policy summary

Policy name	Type	Attached as
AmazonEC2FullAccess	AWS managed	Permissions policy
AmazonSNSFullAccess	AWS managed	Permissions policy
CloudWatchAgentServerPolicy	AWS managed	Permissions policy

Add tags - optional

Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.

No tags associated with the resource.

[Add new tag](#)

Step 1: Select trusted entity

Step 2: Add permissions

Step 3: Name, review, and create

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.

Ec2-SelfHealing

Maximum 64 characters. Use alphanumeric and '+', '@', '-' characters.

Description
Add a short explanation for this role.

Allows EC2 instances to call AWS services on your behalf.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: '_', '=', '@', '/', '!', '\$', '%', '^', '&', '*', '~', '(', ')', '[', ']', '{', '}', '"', ''', '\', '`', 'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', 'k', 'l', 'm', 'n', 'o', 'p', 'q', 'r', 's', 't', 'u', 'v', 'w', 'x', 'y', 'z', '{', '|', '}', '~', '', '€', '', '‚', 'ƒ', '„', '…', '†', '‡', 'ˆ', '‰', 'Š', '‹', 'Œ', '', 'Ž', '', '', '‘', '’', '“', '”', '•', '–', '—', '˜', '™', 'š', '›', 'œ', '', 'ž', 'Ÿ', ' ', '¡', '¢', '£', '¤', '¥', '¦', '§', '¨', '©', 'ª', '«', '¬', '­', '®', '¯', '°', '±', '²', '³', '´', 'µ', '¶', '·', '¸', '¹', 'º', '»', '¼', '½', '¾', '¿', 'À', 'Á', 'Â', 'Ã', 'Ä', 'Å', 'Æ', 'Ç', 'È', 'É', 'Ê', 'Ë', 'Ì', 'Í', 'Î', 'Ï', 'Ð', 'Ñ', 'Ò', 'Ó', 'Ô', 'Õ', 'Ö', '×', 'Ø', 'Ù', 'Ú', 'Û', 'Ü', 'Ý', 'Þ', 'ß', 'à', 'á', 'â', 'ã', 'ä', 'å', 'æ', 'ç', 'è', 'é', 'ê', 'ë', 'ì', 'í', 'î', 'ï', 'ð', 'ñ', 'ò', 'ó', 'ô', 'õ', 'ö', '÷', 'ø', 'ù', 'ú', 'û', 'ü', 'ý', 'þ', 'ÿ'.

Step 1: Select trusted entities

Trust policy

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {

```

3. Create Launch Template

- Go to EC2 - Launch Templates - Create launch template
- name: *Self-Healing-Template*

- AMI:
 - Amazon Linux 2
- Instance type:
 - t2.micro
- Key pair:
 - Select or create one
- Network settings:
 - Allow HTTP (port 80) and SSH (port 22)
- IAM instance profile:
 - Select *EC2-SelfHealing*
- User Data (Install Apache)

```
#!/bin/bash
```

```
yum update -y
```

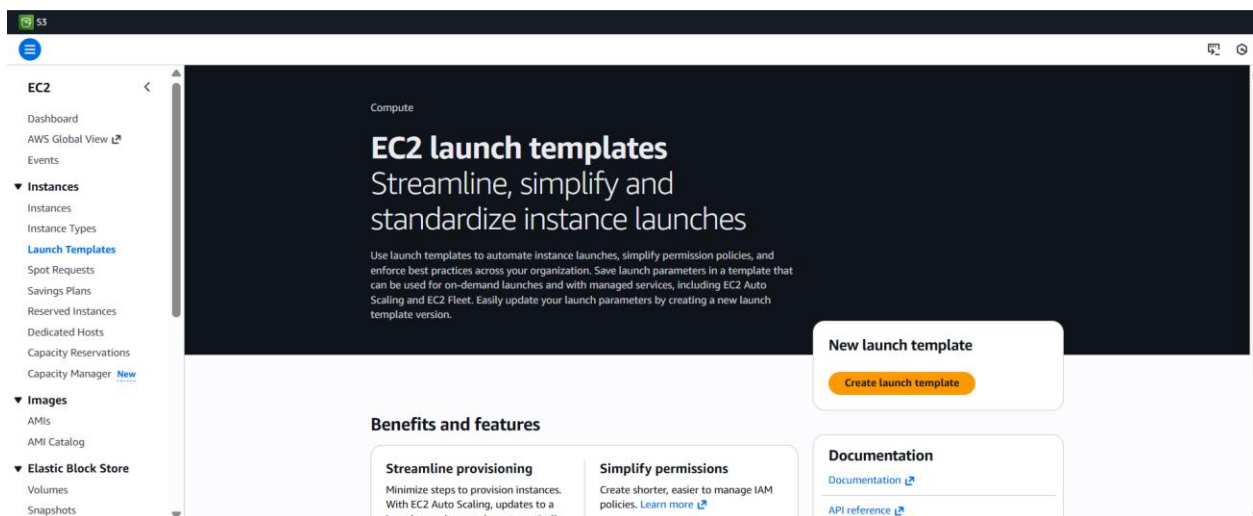
```
yum install httpd -y
```

```
systemctl start httpd
```

```
systemctl enable httpd
```

```
echo "Self Healing EC2 is running" > /var/www/html/index.html
```

- Click Create launch template



RecentsQuick Start

Don't include in launch template

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

Search

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

ami-0ced6a024bb18ff2e (64-bit (x86), uefi-preferred) / ami-01605b62a13170a53 (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.10.20260105.0 x86_64 HVM kernel-6.1

Architecture

64-bit (x86)

Boot mode

uefi-preferred

AMI ID

ami-0ced6a024bb18ff2e

Publish Date

2026-01-02

Username

ec2-user

Verified provider

EC2 > Launch templates > Create launch template

Create launch template

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - required

self-healing-template

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '*', '/', '@'.

Template version description

A prod webserver for MyApp

Max 255 chars

Auto Scaling guidance

Select this if you intend to use this template with EC2 Auto Scaling

☐ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

Template tags

Source template

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

Summary

Software Image (AMI)

Virtual server type (instance type)

Firewall (security group)

Storage (volumes)

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance.

Cancel

Create launch template

Instance type

Info | Get advice

Advanced

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.017 USD per Hour On-Demand RHEL base pricing: 0.0268 USD per Hour On-Demand Linux base pricing: 0.0124 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0142 USD per Hour On-Demand SUSE base pricing: 0.0124 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login)

Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

mykey

Create new key pair

1/2,51.0.0/16

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

Remove

Type

Info

ssh

Protocol

Info

TCP

Port range

Info

22

Source type

Info

Anywhere

Source

Info

Q Add CIDR, prefix list or security group

0.0.0.0/0

Description - optional

Info

e.g. SSH for admin desktop

▼ Security group rule 2 (TCP, 80, 0.0.0.0/0)

Remove

Type

Info

HTTP

Protocol

Info

TCP

Port range

Info

80

Source type

Info

Anywhere

Source

Info

Q Add CIDR, prefix list or security group

0.0.0.0/0

Description - optional

Info

e.g. SSH for admin desktop

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

▼ Summary

Software image (AMI)

Amazon Linux 2023 AMI 2023.10...[read more](#)

ami-0ced6a024bb18ff2e

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance.

Cancel

Create launch template

▼ Advanced details

Info

IAM instance profile

Info

Ec2-SelfHealing

arn:aws:iam::989615686095:instance-profile/Ec2-SelfHealing

Create new IAM profile

Hostname type

Info

Don't include in launch template

DNS Hostname

Info

☐ Enable resource-based IPv4 (A record) DNS requests
 ☐ Enable resource-based IPv6 (AAAA record) DNS requests

Instance auto-recovery

Info

Don't include in launch template

Shutdown behavior

Info

Don't include in launch template

User data - optional

Info

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash
yum update -y
yum install httpd -y
systemctl start httpd
systemctl enable httpd
echo "Self Healing EC2 is running" > /var/www/html/index.html
```

☐ User data has already been base64 encoded

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance.

Cancel

Create launch template

4. Create Auto Scaling Group (ASG)

- Go to EC2 - Auto Scaling Groups - Create
- Select launch template:
 - self-healing-template
- Choose VPC and subnet
- Group size:
 - Desired: 1
 - Minimum: 1
 - Maximum: 1
- Health checks:

- Type: EC2

☰ [EC2](#) > [Auto Scaling groups](#) > Create Auto Scaling group

☰ [EC2](#) > [Auto Scaling groups](#) > Create Auto Scaling group

EC2 > Auto Scaling groups > Create Auto Scaling group

EC2 > Auto Scaling groups > Create Auto Scaling group

Disabled Disabled Disabled

Capacity Reservation preference

Preference Default	Capacity Reservation IDs -	Resource Groups -
-----------------------	-------------------------------	----------------------

Step 5: Add notifications [Edit](#)

Notifications
No notifications

Step 6: Add tags [Edit](#)

Tags (0)

Key	Value	Tag new instances
No tags		

[Preview code](#) [Cancel](#) [Previous](#) [Create Auto Scaling group](#)

EC2 > Instances

Instances (1) [Info](#)

Find Instance by attribute or tag (case-sensitive) [All states](#)

Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

☐	Name	Zuno_UUID	Instance ID	Instance state	Instance type	Status check	Alarm status
☐			i-083de94427c31f26d	Running	t2.micro	2/2 checks passed	View alarms

Configure Cloudwatch

5. Go to SNS - Topics - Create topic

- Type:
 - Standard
- Name:
 - ec2-self-healing-alerts
- Create topic
- Add Email Subscription
- Click Create subscription
- Protocol:
 - Email
- Endpoint:
 - email address
- Create subscription
- Confirm from your email inbox

Amazon SNS > Subscriptions > Create subscription

Create subscription

Details

Topic ARN

arn:aws:sns:ap-south-1:989615686095:Ec2-SelfHealing-Aerts

Protocol

The type of endpoint to subscribe

Email

Endpoint

An email address that can receive notifications from Amazon SNS.

test@example.com

After your subscription is created, you must confirm it. [Info](#)

► Subscription filter policy - optional [Info](#)

This policy filters the messages that a subscriber receives.

6. Create Lambda Function
 - Go to Lambda - Create function
 - Choose:
 - Author from scratch
 - Function name:
 - EC2-SelfHealing-Lambda
 - Runtime:
 - Python 3.9
 - Execution role:

- Create new role with basic permissions
- Go to IAM - Roles -Lambda role
- Attach:
 - *AmazonEC2FullAccess*
 - *AmazonAutoScalingFullAccess*
 - *AmazonSNSFullAccess*
 - *CloudWatchLogsFullAccess*

Logic Code(Get from Chatgpt)

```
import boto3

def lambda_handler(event, context):

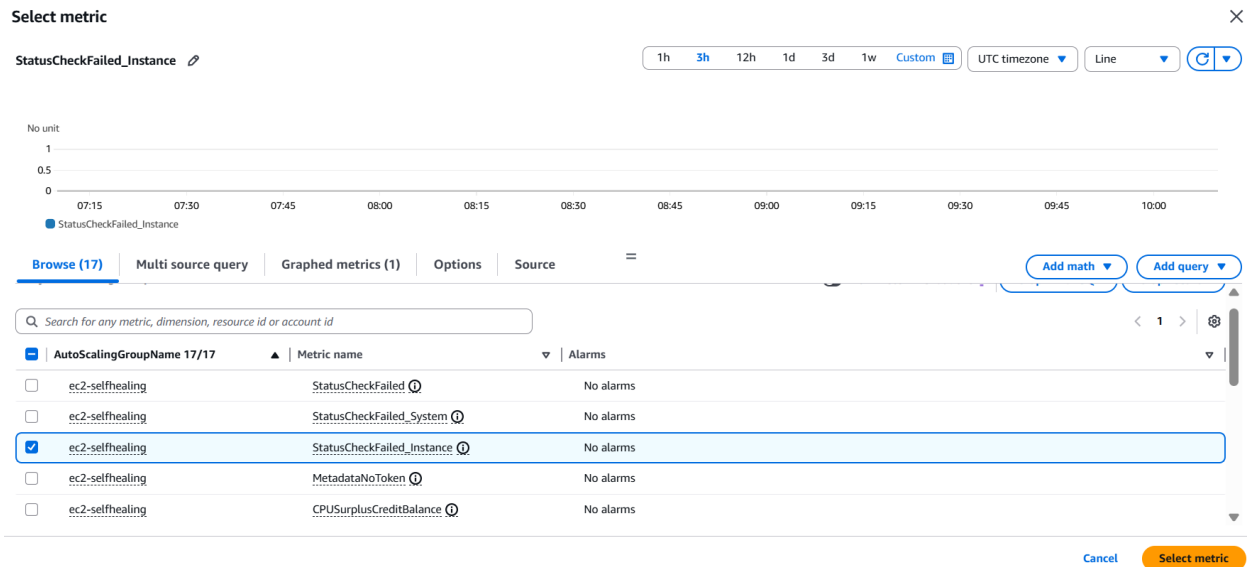
    sns = boto3.client('sns')

    topic_arn = "Ec2-SelfHealing-Aerts"

    sns.publish(
        TopicArn=topic_arn,
        Subject="EC2 Failure Detected",
        Message="CloudWatch detected EC2 failure. Auto Scaling will
replace the instance."
    )

    return {
        'statusCode': 200,
        'body': 'Notification sent'
    }
```


- EC2 – By Auto Scaling Group
- Choose : *StatusCheckFailed_Instance*
- Conditions:
 - Threshold: Greater than 0
 - Period: 1 minute
 - Evaluation periods: 1
- Alarm name: EC2-Failure-Alarm



CloudWatch > Alarms > Create alarm 🔍

0.5
0
0

07:30 08:00 08:30 09:00 09:30 10:00

● StatusCheckFailed_Instance

AutoScalingGroupName

ec2-selfhealing

Statistic

Average ✕

Period

1 minute ▼

Conditions

Threshold type

☒ Static
Use a value as a threshold

☐ Anomaly detection
Use a band as a threshold

Whenever StatusCheckFailed_Instance is...
Define the alarm condition.

☒ Greater
> threshold

☐ Greater/Equal
≥ threshold

☐ Lower/Equal
≤ threshold

☐ Lower
< threshold

than...
Define the threshold value.

0

CloudWatch > Alarms > Create alarm

Step 1: Specify metric and conditions
Step 2: **Configure actions**
Step 3: Add alarm details
Step 4: Preview and create

Configure actions

Notification

Alarm state trigger
Define the alarm state that will trigger this action.

☒ **In alarm**
The metric or expression is outside of the defined threshold.

☐ **OK**
The metric or expression is within the defined threshold.

☐ **Insufficient data**
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☒ **Select an existing SNS topic**
☐ Create new topic
☐ Use topic ARN to notify other accounts

Send a notification to...

arn:aws:sns:ap-south-1:989615686095:Ec2-SelfHealing-Aerts

Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.

Email (endpoints)
ajit.jadhav@acc.ltd - [View in SNS Console](#)

[Add notification](#)

Add lambda function

CloudWatch > Alarms > Ec2-failure-alarm > Edit

Lambda action

Alarm state trigger
Define the alarm state that will trigger this action.

☒ **In alarm**
The metric or expression is outside of the defined threshold.

☐ **OK**
The metric or expression is within the defined threshold.

☐ **Insufficient data**
The alarm has just started or not enough data is available.

Function Type
Select the type of Lambda Function

☒ **Select Lambda Function from the signed in account**
☐ Enter Function ARN for cross account function

Choose a function

Ec2-Self_healing-Lambda

Existing Lambda Functions on the signed-in account, enter the function ARN for cross account functions

Configure version/alias

[Add Lambda action](#)

Auto Scaling action

[Add Auto Scaling action](#)

Test :

- Go to EC2 - Instances
- Select the EC2 instance created by ASG
- Click Instance state - Stop
- Do not restart it

Observe

While setting up Realize that the process o manually doing to instances like stop/terminate Is not detected by Alarm,
Because Alram only detect internal process like OS crash, Status Unhealthy etc

So now we will set a Eventbridge rule to check changing of states of instances

So, thus will detect our instance state and trigger our lambda function.

Create EventBridge Rule

Amazon EventBridge > Rules > Create rule

visual rule builder opt in
If you want to create a scheduled rule use the [scheduled rule builder](#)

Events Targets

Keywords, e.g. "Lambda", "SQS" or "S3"

MOST POPULAR

- EventBridge event bus
- Lambda function
- SQS queue
- SNS topic
- Step Functions state machine
- EventBridge API Destination
- Firehose stream

ALL

Build Configure

Cancel Reset Create

Target

Target configuration - Lambda function [Learn more](#)

Target location

☒ Target in this account ☐ Target in another AWS account

Function

Ec2-Self_healing-Lambda [Create Lambda function](#)

Configure version/alias

Permissions

☒ Use execution role (recommended)

Execution role

EventBridge and AWS Identity and Access Management [Learn more](#)

☒ Create a new role for this specific resource ☐ Use existing role

Role name

Amazon EventBridge

Dashboard

Developer resources

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Buses

- Event buses
- Rules [Updated](#)
- Global endpoints
- Archives
- Replays

Pipes

- Pipes

Scheduler

- Schedules
- Schedule groups

Integration

Partner event sources

CloudShell Feedback Console Mobile App

Application Integration

Amazon EventBridge

A serverless service for building event-driven applications

Amazon EventBridge is a serverless service that uses events to connect application components together, making it easier for developers to build scalable event-driven applications.

Get started

- ☒ **EventBridge Rule with event pattern**
A rule matches incoming events and sends them to targets for processing.
- ☐ **EventBridge Scheduled rule**
A rule that will invoke a target at a scheduled time.
- ☐ **EventBridge Pipe**
A pipe connects an event source to a target with optional filtering and enrichment.
- ☐ **EventBridge Schedule**
A schedule invokes a target one-time or at regular intervals defined by a cron or rate expression.
- ☐ **EventBridge Schema registry**
Schema registries collect and organize schemas.

[Create rule](#)

Pricing

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Amazon EventBridge > Rules > EC2_Failure_Notifications

Amazon EventBridge < EC2_Failure_Notifications [Edit] [Disable] [Delete] [CloudFormation Template]

Dashboard

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▼ Integration

- Partner event sources

Rule details Info

Rule name EC2_Failure_Notifications	Status Enabled	Event bus name default	Type Standard
Description	Rule ARN arn:aws:events:us-east-1:123456789012:rule/EC2_Failure_Notifications	Event bus ARN arn:aws:events:us-east-1:123456789012:bus/default	

Event pattern Targets Monitoring Tags

Event pattern Info [Edit]

```
1 {
2   "source": ["aws-ec2"],
3   "detail-type": ["EC2 Instance State-change Notification"],
4   "detail": {
5     "state": ["stopped", "terminated"]
6   }
7 }
```

[Copy]

Now we stop our running instance

Instances (1/5) Info Last updated 10 minutes ago [Connect] [Instance state] [Actions] [Launch instances]

Find Instance by attribute or tag (case-sensitive) [All states]

	Name	Zuno_UUID	Instance ID	Instance state	Instance type	Status check	Alarm status
☐			i-0e748d9f06c2c2146	Terminated	t2.micro	-	View alarms +
☐			i-0d8ea1d1be9e7c91c	Terminated	t2.micro	-	View alarms +
☒			i-076cab583e593dd40	Shutting-d...	t2.micro	Initializing	View alarms +
☐			i-0bd689f32228db3d1	Terminated	t2.micro	-	View alarms +

i-076cab583e593dd40

Notification Via SNS

AN AWS Notifications<no-reply@sns.amazonaws.com> [Smiley] [Reply] [Forward] [Share] [More]

To: Ajit Jadhav Wed 1/14/2026 6:26 PM

CloudWatch detected EC2 failure. Auto Scaling will replace the instance.

--

If you wish to stop receiving notifications from this topic, please click or visit the link below to unsubscribe:
<https://sns.ap-south-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:ap-south-1:989615686095:Ec2-SelfHealing-Aerts:69612172-0aa0-46bc-9dc3-5207d89fbf99&Endpoint=ajit.jadhav@acc.ltd>

Please do not reply directly to this email. If you have any questions or comments regarding this email, please contact us at <https://aws.amazon.com/support>

[Reply] [Forward]

New instance Launched By ASG

Instances (1/5) [Info](#)

Find Instance by attribute or tag (case-sensitive)

All states

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

< 1 >

☐	Name	Zuno_UUID	Instance ID	Instance state	Instance type	Status check	Alarm status
☐			i-0e748d9f06c2c2146	Terminated	t2.micro	-	View alarms +
☒			i-076cab583e593dd40	Terminated	t2.micro	-	View alarms +
☐			i-0ab458850cd6d67d0	Running	t2.micro	Initializing	View alarms +
☐			i-0bd689f32228db3d1	Terminated	t2.micro	-	View alarms +
☐			i-07c42d4739e1371f3	Terminated	t2.micro	-	View alarms +

i-076cab583e593dd40

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags